Bianca Bragg

HW # 8

7A.1. Pressure Rise in a Sudden Enlargement

- momentum balance -

$$\vec{F}_{f\to s} = (v_1 w_1 - \overset{\to 0}{v_2 w_2}) + (\underbrace{p_1 S_1 - p_2 S_2}_{\parallel\; p_1 (S_1 - S_2)})$$

$$\Rightarrow -p_1(S_2 - S_1) = \rho v_2 S_2 (v_1 - v_2) + (p_1 S_1 - p_2 S_2)$$

since $\begin{cases} w_1 = w_2 \\ \rho v_1 S_1 = \rho v_2 S_2 \\ v_1 / v_2 = S_2 / S_1 = 1/\beta \end{cases}$, $\beta = S_1 / S_2$

Then, solving for Δp:

$$p_2 - p_1 = \rho v_2^2 (1/\beta - 1).$$

Solving for β:

$$\beta = \frac{S_1}{S_2} = \frac{D_1^2}{D_2^2} = \frac{5^2}{9^2} = \frac{25}{81}$$

$$= 0.309.$$

Solving for V_2:

$$V_2 = \frac{Q}{S_2} = \frac{(450 \text{ gal}/60 \text{ s})(0.13368 \text{ ft}^3/\text{gal})}{(\pi/4)(25/144) \text{ ft}^2}$$

$$= 7.35 \text{ ft/s}.$$

Finally, we can compute Δp

$$p_2 - p_1 = \rho V_2^2 (1/\beta - 1)$$

$$= (63 \text{ lb}_m/\text{ft}^2)(7.35 \text{ ft/s})^2 (1/0.309 - 1)$$

$$= 7{,}610 \text{ poundals}/\text{ft}^2$$

$$\shortparallel$$

$$1.64 \text{ psi}$$

7A.2 Pumping a Hydrochloric Acid Solution.

From.

$$\Delta\left(\frac{1}{2}\frac{\langle v^3\rangle}{\langle v\rangle}\right) + g\Delta h + \int_1^2 \frac{1}{\rho}\,dp = \hat{W}_m - \hat{E}_v$$

$$|||$$

$$\frac{1}{2}\left(v_2^2 - v_1^2\right) + g\left(z_2 - z_1\right) + \int_{p_1}^{p_2} \frac{1}{\rho}\,dp =$$

$$= \hat{W}_m - \sum_i \left(\frac{1}{2} v^2 \frac{L}{R_h} f\right)_i - \sum_i \left(\frac{1}{2} v^2 e_v\right)_i$$

$$|||$$

$$\hat{W}_m = \frac{p_2 - p_1}{\rho} + \left(\frac{1}{2} v^2 \frac{L}{R_h} f\right) + \left(\frac{1}{2} v_2^2\right),$$

where.

$$\frac{p_2 - p_1}{\rho} = \frac{(3.0)\ \text{atm}\ (6.8087\times10^4\ lb_m/ft\cdot s^2\cdot atm)}{(62.4\ lb_m/ft^3)}$$

$$= 3273\ ft^2/s^2$$

and also:

$$\frac{1}{2} v^2 \frac{L}{R_h} f \equiv \frac{1}{2} v^2 \frac{L}{D/4} f =$$

$$= \frac{1}{2} (2.30 \, ft/s)^2 \cdot \frac{300 \, ft}{(0.5/12) \, ft} (0.0048) =$$

$$= 92 \, ft^2/sec^2$$

and:

$$\frac{1}{2} v_2^2 = \frac{1}{2} (2.30 \, ft/s)^2 =$$

$$= 2.6 \, ft^2/s^2$$

Thus, calculating the work/unit mass fluid:

$$\hat{W}_m = (3273) + (92) + (2.6)$$

$$= 3368 \, ft^2/s^2$$

$$W = \hat{W}_m \cdot w, \text{ where}$$

$$w = \pi R^2 \rho \langle v \rangle$$

$$= \pi \left(\frac{2^2}{12^2} ft^2\right)(62.4 \, lb_m/ft^3)(2.3 \, ft/s)$$

$$= 12.5 \, lb_m/s$$

Finally,

$$W = \hat{W} w$$

$$= (3368\ ft^2/s^2)(12.5\ lb_m/s)$$

$$= (4.22 \times 10^4\ ft\ poundals/s)(3600\ s/hr.) \times (1.5698 \times 10^{-8}\ hp\text{-}hr/ft\ poundal)$$

$$= 2.4\ hp$$

$$= 1.8\ kW.$$

7A.4 Incompressible Flow in an Annulus.

$T = 60°F$
$L = 20.3\ ft$
$w = 241\ gal/min$
$R_o = 7\ in.$
$R_i = 3\ in.$
$\Delta h = 5\ ft.$

(i) $$W = \left(241\ \frac{gal}{min}\right)\left(\frac{231}{1728}\ \frac{ft^3}{gal}\right)\left(62.4\ \frac{lb_m}{ft^3}\right)\left(\frac{min}{60s}\right)$$

$$= 33.5\ lb_m/s$$

(ii) $$R_h = S/Z, \quad S = \pi(R_o^2 - R_i^2), \quad Z = 2\pi(R_o + R_i) \Rightarrow$$

$$\Rightarrow R_h = \frac{1}{2}(R_o - R_i) =$$

(iii) $$\langle v \rangle = \frac{w}{\rho S} = \frac{w}{\rho(\pi(R_o^2 - R_i^2))} =$$

$$= \frac{33.5\ lb_m/s}{(64\ lb_m/ft^3)\,\pi\left((7/12)^2 - (3/12)^2\right)ft^2} =$$

$$= 0.615\ ft/s$$

(iv) $Re = \frac{4 R_h \langle v \rangle \rho}{\mu}$

$$= \frac{2(R_o - R_i) w}{\pi (R_o^2 - R_i^2) \mu} = \frac{2 \cancel{(R_o - R_i)} w}{\pi \cancel{(R_o - R_i)} (R_o + R_i) \mu} =$$

$$= \frac{2 w}{\pi (R_o + R_i) \mu} = \frac{2 (33.5 \ lb_m / s)}{\pi (7/12 + 3/12) ft \left(1.14 \times 0.000672 \ \frac{lb_m}{ft \cdot s}\right)}$$

$$= 3.34 \times 10^4 \Rightarrow \text{turbulent flow}$$

$$\Rightarrow f = 0.0059$$
(fig 6.2-2)

(v) $\hat{W} = g \Delta h + \frac{1}{2} \langle v \rangle^2 \frac{L}{R_h} f$

$$= g (h_2 - h_1) + \langle v \rangle^2 \frac{L}{R_o - R_i} f$$

$$= (32.2 \ ft/s^2)(5 - 0) + (0.65 \ ft/s)^2 \left(\frac{20.3}{7/12 - 3/12}\right) (0.0059) =$$

$$= 161 + 0.175 = 161.1 \ ft^2/s^2$$

Finally,

(vi) $W = w\hat{W}$

$= (33.5\ lb_m/s) \cdot (161.1\ ft^2/s^2) =$

$= 1.9 \times 10^7$ ft. poundals/hr

$\equiv$

0.31 hp

$\equiv$

0.23 kw.

7B.1 Velocity Averages from the 1/7-Power Law.

(i) $$\frac{\langle \bar{v}_z \rangle}{v_{max}} = \frac{\int_0^{2\pi}\int_0^R \left(1-\frac{r}{R}\right)^{1/7} r\,dr\,d\theta}{\int_0^{2\pi}\int_0^R r\,dr\,d\theta} =$$

$$= 2\int_0^1 (1-\xi)^{1/7}\xi\,d\xi$$

$$= 2\int_0^1 x(1-x^7)(7x^6)\,dx$$

$$= \frac{60}{14}\left(\frac{15}{8} - \frac{8}{15}\right) = \frac{49}{60}$$

(ii) $$\frac{\langle \bar{v}_z^2 \rangle}{v^2_{max}} = \frac{\int_0^{2\pi}\int_0^R \left(1-\frac{r}{R}\right)^{2/7} r\,dr\,d\theta}{\int_0^{2\pi}\int_0^R r\,dr\,d\theta} =$$

$$= 2\int_0^1 (1-\xi)^{2/7}\xi\,d\xi$$

$$= 2\int_0^1 x(1-x^{7/2})\left(\frac{7}{2}x^{5/2}\right)dx$$

$$= \frac{72}{7}\left(\frac{8}{9} - \frac{9}{8}\right) = \frac{49}{72}.$$

Thus,

$$\frac{\langle \bar{v}_2^{\,2} \rangle}{\langle \bar{v}_2 \rangle^2} = \left(\frac{49}{72}\right)\left(\frac{60^2}{49^2}\right) = \frac{50}{49}$$

$$\text{(iii)} \; \frac{\langle \bar{v}_2^{\,3} \rangle}{v^3_{max}} = \frac{\int_0^{2\pi}\int_0^R \left(1-\frac{r}{R}\right)^{3/4} r\,dr\,d\theta}{\int_0^{2\pi}\int_0^R r\,dr\,d\theta}$$

$$= 2\int_0^1 (1-\xi)^{3/4}\,\xi\,d\xi$$

$$= 2\int_0^1 x(1-x^{7/3})\left(\frac{7}{3}x^{4/3}\right)dx$$

$$= \frac{14}{3}\left(\frac{3}{10}-\frac{3}{17}\right) = \frac{49}{85}.$$

Then,

$$\frac{\langle \bar{v}_2^{\,3} \rangle}{\langle \bar{v}_2 \rangle^3} = \left(\frac{49}{85}\right)\left(\frac{60}{49}\right)^3$$

$$= \frac{43200}{40817}.$$